

Lab Goal : This lab was designed to teach you more about linked lists. In this lab the list is stored in an instance variable and will be accessed by the instance methods, unlike last lab where the list was created in the main method and passed in as a parameter to the static methods.

Lab Description : Use the `ListNode` class given below to write some basic Linked List methods. In this implementation only a reference to the first node is stored in a reference variable. You access the other blocks by iterating to them from the first node.

ListNode – stores a value and a reference to the next node

```
public class ListNode implements Linkable
{
    private Comparable listNodeValue;
    private ListNode nextListNode;

    public ListNode(Comparable value, ListNode next){
        listNodeValue=value;
        nextListNode=next;
    }

    public Comparable getValue(){
        return listNodeValue;
    }

    public ListNode getNext(){
        return nextListNode;
    }

    public void setValue(Comparable value){
        listNodeValue = value;
    }

    public void setNext(Linkable next){
        nextListNode = (ListNode)next;
    }
}
```

Sample Data :

See the main of runner.

Sample Output :

Original list values: go on at 34 2.1 -a-2-1 up over
num nodes = 8

List values after calling **nodeCount**: go on at 34 2.1 -a-2-1 up over

List values after calling **doubleFirst**: go go on at 34 2.1 -a-2-1 up over

List values after calling **doubleLast**: go go on at 34 2.1 -a-2-1 up over over

List values after calling **skipEveryOther**: go on 34 -a-2-1 over

List values after calling **removeXthNode(2)**: go 34 over

List values after calling **setXthNode(2,one)**: go one over

List values after calling **removeXthNode(1)**: